Product Specifications

Item	SPAD-050-C-B-FC	SPAD-050-C-R-FC
Electrical Specifications		
Electrical safety class	Class III	
Power supply	9 V DC (2.1/5.5 mm) or 5 V DC via USB Type-C	
Maximum current consumption	0.6 A	
Maximum power consumption	3 W	
Signal and Optical Specifications		
Sensor type	Single Photon Avalanche Diode (SPAD)	
Photosensitive area	50 µm	
Spectral response range	370 nm – 900 nm	400 nm – 1000 nm
Peak sensitivity	450 nm	630 nm
Maximum incident optical power	50 µW	
Maximum count rate	3.5 Mcps	3 Mcps
Dark count rate (typical)	7 cps	20 cps
Optical input	1x FC/PC multimode fiber connector	
Outputs	1x SMA (LVTTL) 1x USB Type-C (LVDS)	
Output pulse amplitude	SMA: 2.5 V LVTTL (50 Ω) USB Type-C: LVDS	
Output pulse width	10 ns	
Timing jitter	65 ps RMS	
Impedance	50 Ω	
Physical Specifications		
Dimensions (L x W x H)	127 mm x 77 mm x 40 mm	
Weight	344 g	
Protection rating	IP40	
Environmental Specifications		
Operating temperature	0°C to +40°C	
Storage temperature	-20°C to +70°C	
Relative humidity	Up to 80% non-condensing	